SAP PP overview:

SAP PP converts demand into a feasible production Plan and then into actual manufactured

PP modules (sub): 8

PP-BD

PP-MRP - Materials

PP-CRP -> Capacity

PP-SFC -> Manages Production Execution

PP-PE -> Production execution

PP-REM -> Repetitive manufacturing

PP-KANBAN

PP-DM

Manufacturing types: 4

- Discrete - EX- Automobile, Electronics
- Process - EX- Pharma, Food
  - -> uses - BoM, Routing, Work center
  - -> uses - Master Recipe, Resource
- Repetitive - EX - Consumer Goods, Fasteners

↓

High volume, No individual production orders.

KANBAN - Pull-based replenishment uses control cycle.

Production Strategies : 4

MTS - Forecast, customer buys from stock
MTO - Production strat after order
ATO - Final assembly after order
ETO - Design happens after customer order

PP organizational units : 6

Plant
↓
BOM, Routing, Stock, MRP settings, work centers
Storage location
↓
Raw material store, Finished Good store
MRP area
↓
Planning subdivision within a plant.
MRP Controller
↓
Responsible for planning materials
Production Scheduler
↓
" " production & execution
Work Center
↓

Machine, Production line or labor group

PP integration: 5

PP + MM - Generates material requis

PP + SD

PP + QM

PP + PM

PP + FI/CO

MD04

4 Important People in PP:

Production Planner

Production Supervisor

Machine operator

Plant Controller

Important Transactions: 4

Master Data - MD61, MD04 MM01, CS01, CR01, CA

Planning - C007, C011N MD61, MD04

Execution - C001, C011N

Monitoring - C024, MB51

SAP PP - Helps the organizations to Plan, manufacture and control the process of production.

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Integrates with - MM, SD, QM, WM

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Objectives - P.P, M.P, C.P, P.E, I.C

Material master - central repository of information about a material.

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Types - ROH, HALB, FERT, HAWA, VERP

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Views - Basic Data, MRP views, Purchasing view, Accounting View, Storage View, work scheduling view.

Billing of material - list of materials required to manufacture a product.

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Required - MRP, Production order creation cost calculation, component explosion

Routing -> Routing defines the sequence of the operations required to manufacture a product.

Routing contains - operations, work centers, control keys, standard keys values.

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Why it is required - scheduling, capacity planning, costing, Production execution.

Production version: this links BOM, Material and Routing into a valid production process.

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Why - Must know which BOM to use and which Routing to use for production

Demand management - It is used to plan the future demand for products.

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Main output - PIR

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Better planning, Material availability, capacity planning, Reduce Shortages

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Benefits

SOP - High level planning process that balances Sales Forecast, Production capacity Inventory levels.

Control key - Determines how an operation is processed in a routing or PO

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Purpose - Be scheduled, Be costed.

PP01 - Sch + capa plan cost config

Standard value key - defines standard values that can be entered in routing operations

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Purpose - links routing values with work center formulas

purpose - converts business forecasts into production plans.

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Flow - Sales Forecast

↓

SOP

↓

D.M

↓

MRP

↓

Production.

LYP - Simulation planning tool used to analyze future scenarios. without impacting actual planning data.

Work center : represents location or resource where production operations are carried out.

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Functions - scheduling, capacity planning, costing, production execution

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views - B.D, D.V, capacity, sales, scheduli

ERP - is a software that integrates all business processes into a single system.

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Helps - Finance, Procurement, sales, Inventory, production, Human Resources.

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Benefits - Real-time information, Better decision-making, Reduced data duplication, Improved Productivity

Modules in ERP - PP, MM, SD, FI, CO, QM, PM, WM, HCM

SAP - systems, Applications and Products in Data processing.

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Purpose - Manage business processes, Integrated departments, Improve efficiency, make data-driven decisions

SAP S/4HANA - Lastest ERP suite

S = Simple

4 = Fourth Generation

HANA = High-Performance Analytic Appliance

It runs HANA Database
HANA - in memory database
stores the data in memory (RAM)
↓
Benefits - Faster processing, Real-time reporting, Faster mrp runs, Better analytics
Master Data - Data that remains relatively stable and is used repeatedly.
↓
M.M, BoM, Routing, W.C, P.V
Transaltional data - created by day-to-day bussieness needs
↓
Example - P.O, Goods Receipt, Goods issue, P.O, S.O
structure - client → company code → Plant → storage location → Hierarchy
End-to-End manfactiuring Flow -
Demand → PIR → MRP → Planned order → Production order → Goodes Issue → Production confirmation → Goods Receipt